



Proposed CEA Link — December 2026

Weekday Lates with weeks 14–17 kept in order and ON THEIR OWN LINES, week 13 moved away, and the evening gap filled

Prepared 22 September 2026 · built for the December 2026 timetable and assessed by the workspace's own rule modules · figures on this page are computed, not typed

PROPOSAL

Weekday Lates 4

Weeks 14–17 kept in order, on their own line numbers

CODE

WL4-24-F7

FINGERPRINT

f0d403d6

BUILT FROM

duty table — · seed — · 24 lines · 4 cover weeks

LINEAGE

Weekday Lates (WL-24-EXT · a52ec588) with the 08:30–17:00 turns outside the protected weeks re-timed into the evening, weeks 14–17 held in order at their own line numbers, week 13 held as written but free to move, and the rotation then searched fatigue-first. Shares its starting grid, duty table and coverage curve with Weekday Lates 3 (WL3-24-F7 · a6234195), which kept the same five weeks but moved all of them.

✓ **24 lines** designed

✓ **35h 00m** a week — exactly the contract

✓ **0** hard-limit breaches

✓ **0** rests under 12h

⚠ **1** fatigue factor present (today: 5)

13 of 19

shift times are today's

6 times are new; the rest are turns people already work

7 days

longest run of worked days

today's link reaches 9; Chiltern's limit is 13

10 of 20

working weeks are one turn

same clock time all week, as 7 of today's 16 are

6 in 24

full weekends off

25% of the rotation, against 20% today (4 in 20)

4 cover weeks

lines 1, 7, 12, 17

gaps of 8, 6, 5, 5 around the wheel — inherited from Weekday Lates, not re-spread

2h 34m a week

typical week-to-week move

how far the working day shifts each week — today's link moves 4h 0m a week

What this is

A proposal for the CEA link once the December 2026 timetable arrives, for a team of 24 rather than 20. The brief was to keep the **feel** of the link people work today — a week is one turn, rest days come in pairs, four whole cover weeks spread round the rotation, and the same 06:20 / 08:00 / 14:00 / 15:15 clock times — while meeting the staffing shape agreed for December (four to open every day, three through to the close, four on a Saturday, five still on at 22:00, fourteen on a Saturday, ten on a Sunday) and every rule the link is assessed against: Chiltern's 13-day limit, twelve hours between duties, the exact 35-hour contracted week, and the ORR's good-practice fatigue factors.

It was not drawn by this app. It arrived as a Word table — a proposal from outside the workspace — and was read straight into the app's own shape, then judged by the same modules that judge a design the workspace generated itself: `runDesignChecks`, `weeklyHours`, `assessHardLimits`, `assessFatigue`, `scoreOrder` and `calcHourlyCoverage`. There is no search behind it and nothing to reproduce; what can be checked is every figure on these pages, each computed from the cells on page 6. Paste those cells into Links → Import and the workspace restates all of it.

What it is not. Not a recommendation and not a finished roster: nothing here has been through the roster office or a rep. The 24-line length, the four cover weeks and the December headcounts are owner-relayed figures with no document behind them (evidence class C), and the 13-day limit's policy citation is still outstanding — pages 5 and 6 say so where it applies. It clears every rule the tool can check, which is the floor for being worth discussing.

THE ROTATION

The 24-line link

Sunday to Saturday per line; everyone moves down one line each week and line 24 goes back to line 1. Hours and days at the right, cover beneath.

LINE	SUN	MON	TUE	WED	THU	FRI	SAT	MON-SAT hours	ALL WEEK hours	DAYS WORKED Mon-Sat
1	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—
2	08:30 16:30	RD	RD	06:20 14:20	06:20 14:20	06:20 14:20	06:20 14:20	32h 00m	40h 00m	4
3	RD	RD	08:30 16:30	08:30 16:30	07:00 15:30	07:00 15:30	06:20 14:50	41h 30m	41h 30m	5
4	RD	RD	14:00 22:30	13:30 22:00	14:00 22:30	14:00 22:30	14:30 22:00	41h 30m	41h 30m	5
5	14:30 23:25	RD	RD	13:30 22:00	13:30 22:00	13:30 22:00	13:30 22:00	34h 00m	42h 55m	4
6	13:30 22:00	13:30 22:00	RD	RD	13:30 22:00	13:00 21:00	14:45 23:55	34h 10m	42h 40m	4
7	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—
8	07:15 15:45	06:20 13:45	06:20 13:45	RD	RD	06:20 13:35	08:00 16:00	30h 05m	38h 35m	4
9	07:15 15:45	06:20 13:45	06:20 13:45	06:20 13:45	06:20 13:45	RD	RD	29h 40m	38h 10m	4
10	RD	06:20 14:20	07:00 15:30	07:00 15:30	06:20 14:20	06:20 14:20	06:20 13:45	48h 25m	48h 25m	6
11	RD	RD	06:20 14:20	12:30 21:00	12:30 21:00	12:30 21:00	RD	33h 30m	33h 30m	4
12	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—
13	10:30 19:00	12:30 21:00	12:30 21:00	RD	RD	12:30 21:00	13:30 21:00	33h 00m	41h 30m	4
14	14:30 23:25	14:00 22:30	RD	RD	RD	07:00 15:30	06:20 14:50	25h 30m	34h 25m	3
15	07:15 15:45	06:20 13:45	06:20 13:35	06:20 13:35	06:20 13:35	RD	RD	29h 10m	37h 40m	4
16	RD	08:30 17:00	08:30 17:00	08:30 17:00	RD	RD	RD	25h 30m	25h 30m	3
17	SP	SP	SP	SP	SP	SP	SP	35h 00m	35h 00m	—
18	14:30 23:25	16:25 23:55	RD	RD	13:30 22:00	16:25 23:55	RD	23h 30m	32h 25m	3
19	RD	14:00 22:30	14:00 22:30	14:00 22:30	14:00 22:30	14:00 22:30	RD	42h 30m	42h 30m	5
20	RD	07:00 15:30	07:00 15:30	06:20 13:45	07:00 15:30	06:20 13:45	06:20 14:50	48h 50m	48h 50m	6
21	RD	RD	16:25 23:55	16:25 23:55	16:25 23:55	16:25 23:55	16:25 23:55	37h 30m	37h 30m	5
22	RD	16:25 23:55	16:25 23:55	16:25 23:55	16:25 23:55	16:25 23:55	RD	37h 30m	37h 30m	5
23	RD	16:25 23:55	16:25 23:55	16:25 23:55	16:25 23:55	16:25 23:55	RD	37h 30m	37h 30m	5
24	RD	RD	13:30 22:00	13:30 22:00	13:30 22:00	RD	14:45 23:55	34h 40m	34h 40m	4
Cover	9 SP 4	13 SP 4	15 SP 4	15 SP 4	16 SP 4	16 SP 4	12 SP 4	35h 00m	38h 13m	4.35

Early turns Late turns Cover (spare) week — four duties of seven, any turn Rest day

Mon-Sat hours average 35h 00m over 24 lines, a cover week counted as a contracted week · days worked average over the 20 working lines

What was held, what moved, and what it cost

Four of the five protected weeks never move at all; one does, and that is what makes the rest work

Kept — the shape of a week

MEASURE	TODAY (20)	PROPOSED (24)
Working weeks that are one turn (one clock time Mon–Fri)	7 of 16	10 of 20
Weeks mixing early and late turns	7	2
Rest-day breaks that are two days or more	13 of 17	18 of 21
Single rest days between duties (not a 48h break)	4	3
Days worked in a week (lines × days)	1×3, 3×4, 11×5, 1×6	1×3, 4×4, 13×5, 2×6
Cover (spare) weeks (whole weeks)	lines 1, 7, 12, 17	lines 1, 7, 12, 17
Distinct shift times (all already on today's roster)	18	19
Mon–Sat hours, average week (the contract)	35h 00m	35h 00m
Sunday duties (not contracted; half the working lines)	8 of 16 lines	9 of 20 lines

Changed — the December headcount

PEOPLE ON DUTY	TODAY	PROPOSED	WHY
Monday to Friday	12	15	the contract: 20 working lines × 35h has to be worked somewhere
Saturday	10	12	owner's figure for December, leaning late for events
Sunday	8	9	owner's figure for December
Opening at 06:20 (07:15 Sunday)	4	4	unchanged
Through to the 23:55 close, weekday	2	3	three to close
Through to the close, Saturday	2	4	four on a Saturday
Still on duty at 22:00	4	5	five are enough from 22:00

The duty table, in today's times

TURN	WK	SAT	SUN	WK	SAT	SUN
06:20–13:35 7h 15m · Early	1		1			
06:20–13:45 7h 25m · Early	1		3 +2	1 +1		
06:20–14:00 7h 40m · Early		4		4		
06:20–14:20 8h 00m · Early	2		2	1 +1		
06:20–14:50 8h 30m · Early		3		3		
07:00–15:30 8h 30m · Early			2 +2			
07:15–15:45 8h 30m · Early			3		3	
08:00–16:00 8h 00m · Early				1 +1		
08:00–16:30 8h 30m · Early	2		2			
08:30–16:30 8h 00m · Early			1	1 +1		1
08:30–17:00 8h 30m · Early			1 +1			
10:30–19:00 8h 30m · Early					1 +1	
11:00–19:30 8h 30m · Late	4		4			
12:00–20:00 8h 00m · Late		4		4		
12:30–21:00 8h 30m · Late			2 +2			
13:00–21:00 8h 00m · Late			1	1 +1		-1
13:30–21:00 7h 30m · Late		1		1		
13:30–22:00 8h 30m · Late	1		4 +3	1 +1	1 +1	
14:00–22:30 8h 30m · Late	2		2			
14:30–22:00 7h 30m · Late		2		1 -1		
14:30–23:25 8h 55m · Late			3		3	
14:45–23:55 9h 10m · Late		2		2		
15:15–23:55 8h 40m · Late	2		2			
16:25–23:55 7h 30m · Late			4 +4	1 +1		

Busiest weekday · Saturday · Sunday. The first three columns are what the 20-line link does now; the last three are the proposal. Green cells grew, amber shrank, a struck-through row is a time the proposal does not use.

Why the duties land where they do: Monday to Saturday totals exactly 42,000 minutes, which is 20 × 35h to the minute, so the contract is paid. But unlike a table built day-uniform, **the weekdays here are not equal** — Mon 6,225 · Tue 7,205 · Wed 7,205 · Thu 7,780 · Fri 7,690, and Saturday 5,895. Monday is the lightest day by some margin and Thursday the heaviest; that is a property of the design rather than a fault, but it is the first thing to put to the roster office, because it decides how much cover each day really has.

People on duty, hour by hour

Cover today and proposed against the measured December 2026 timetable (arrivals and departures, weighted by train length). A darker orange hour carries more of the day's traffic.

Monday to Friday window 06:20–23:55

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23
TODAY		4	4	6	6	6	7	7	8	8	8	8	6	6	6	5	5	4	2
PROPOSED		4	6	8	8	8	8	9	10	9	8	9	7	7	7	7	6	5	3
DEC 2026 TRAFFIC	5	69	107	127	122	95	85	75	84	79	87	105	140	138	107	90	73	75	66

Saturday window 06:20–23:55

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23
TODAY		4	4	4	4	4	4	5	6	9	6	6	6	6	6	5	4	2	2
PROPOSED		5	5	6	6	6	6	6	8	10	6	6	6	6	6	6	5	3	3
DEC 2026 TRAFFIC	6	24	58	75	81	92	88	77	64	68	72	84	82	67	64	79	78	62	41

Sunday window 07:15–23:25

	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23
TODAY			3	4	4	4	4	4	5	8	8	5	4	4	4	4	3	3	3
PROPOSED			3	4	4	5	5	5	6	9	9	6	5	5	4	4	4	3	3
DEC 2026 TRAFFIC			15	45	65	71	74	71	61	55	64	69	78	73	75	52	71	65	50

Spare cover is not in these figures — a cover week carries no times, so the rows are a floor. Sunday: 5 December movements fall after the 23:25 finish — the standing question on whether Sunday's window moves; the window is stored per design, so this proposal can be rebuilt to either answer.

Reading it. The proposal raises the whole day rather than reshaping it, which is what the December brief asked for — the station rosters posts (ticket office, gateline, passenger assist) plus break cover, and a post needs someone on it whether the hour carries 23 trains or 5. The one hour the search was allowed to favour is Saturday's 17:00–22:00, for events. The weekday's two peaks are a duty length apart, so cover stays fairly flat between them by arithmetic, not neglect.

The checks sheet

Hard limits first — met or not. Then the ORR good-practice fatigue factors, which report what is present and never pass or fail a design.

Hard limits — a design either meets these or cannot be run

- ✓ **More than 13 consecutive days worked** — longest possible run 7 days (today: 9)
Longest possible run is 7 days, within the 13 configured here from Chiltern practice (origin: the legacy Hidden standard). A spare week is 4 duties of 7 and the roster clerk places them, so this is the worst case the link allows.
Basis: Chiltern roster policy, citation outstanding — legacy Hidden 13-in-14 standard. Configured from Chiltern practice; the policy citation is outstanding, so this is stated as the app states it.
- ✓ **At least 12 hours between duties** — 0 rests under 12h anywhere in the rotation, Saturday-into-Sunday and line-into-line included (today: 0)
The generator refuses a design it cannot repair to this; the search here never produced one.
- ✓ **The contracted week, exactly** — 35h 00m average Mon–Sat over the 24 lines, cover weeks counted as contracted weeks
700h of duty a week across 20 working lines. Sundays (77.25h) sit on top as RDW, as they do today. Individual weeks range 23h 30m–48h 50m; only the average is the contract.

December design figures — the staffing shape agreed for the new timetable

RULE	PROPOSAL
Four on at the open, every day	× 4 weekday · 5 Saturday · 3 Sunday
Three through to the close; four on a Saturday	× 3 weekday · 3 Saturday · 3 Sunday
Five still on duty at 22:00	× 5 weekday · 3 Saturday a 22:00 finish is not "on at 22:00"
Fourteen on a Saturday, ten on a Sunday	× 12 · 9
Cover weeks, evenly spread	× 4 weeks at lines 1, 7, 12, 17 — gaps 8, 6, 5, 5 the December shape asks for five
About 4.2 days a week worked, Mon–Sat	✓ 4.35 over the 20 working lines
No :05 or :10 times except the open and close	✓ none
Late turns slightly shorter than most earlies	× not met — lates 7h30–9h10, earlies 7h15–8h30 the longest late is longer than the longest early

Two things to settle before it is frozen

The seven-day run is the price, and it is structural. The longest working stretch is **7 days** against *Weekday Lates 3's* 6. Week 14 works Fri–Sat and week 15 Sun–Thu, so holding those two in order is seven straight days whatever else moves. It is *not* a threshold breach — FF11 allows 13 consecutive and the “more than 7 consecutive 8h shifts” row needs more than seven — but 6 is what the other proposals reach, and this one cannot. **What holding the line numbers cost.** Of the twelve runs, one placement reached **7 full weekends off** against this design's 6 — but it puts the block on lines 22, 23 and 24 instead of 14, 15 and 16. That candidate is in the table opposite: a weekend against the line numbers, and it is the room's call rather than the search's. **FF19 is at its floor.** Two instances of successive start times varying by more than two hours, down from *Weekday Lates 3's* three. No rotation holding both early and late weeks can reach zero — crossing from earlies to lates is by definition a jump over two hours — and two is the arithmetic floor for this mix of turns, which *Weekday Lates 3's* own notes state. **Weeks 13–17 in full.** If all five must stay in order, MRSF is present at 58.9h and the floor is two factors rather than one; that is a property of the 13–14 join, not a limit of the search.

#

METHOD

How it was built, and how to load it

Reproducible: the same starting grid, seed and objective give the same rotation on any machine

How the order was kept

1 • Weeks 14, 15, 16 and 17 do not move. They sit at lines 14, 15, 16 and 17, in that order, exactly as written in *Weekday Lates* — not a day, not a time, not a line number altered. That is the difference between this proposal and *Weekday Lates 3*, which kept the same five weeks to the letter but scattered them to lines 21, 3, 6, 24 and 1.

2 • Week 13 moves, and only week 13. It keeps its duties exactly as written and goes to line 4. It has to move, and the reason is arithmetic rather than preference: week 13 works Tue–Sat and week 14 Sun–Mon, so back to back they are seven consecutive working days totalling **58.9 hours** against MRSF's 55-hour limit. That window lies wholly inside those two weeks, so no rearrangement of the other twenty-two can break it. Separated, the worst seven-day window is 54.7h and the row clears.

3 • The block's position was enumerated, not searched. Cover weeks are pinned at 1, 7, 12 and 17, and week 17 *is* one of them — a cover week is blank, so “14, 15, 16, 17 in order” means the three working weeks sitting immediately before some cover week. On a 24-line wheel that is four placements and no more, so all four were tried at three seeds each. Twelve runs, every one clearing every threshold; the placement that keeps the weeks on their own line numbers is the one printed here.

4 • Then the rest of the wheel was searched fatigue-first. Every factor present charged over and above its size, three restarts a seed, with only the two coverage-preserving moves — so the cover curve is identical to *Weekday Lates 3*'s, hour for hour, and the ordering choice costs nothing in cover.

Paste it into the workspace

Links → Import, then paste the block on page 8. Line number, then Sunday to Saturday; SP is a cover week. The workspace re-runs every check on these pages from the pasted cells, so nothing here has to be taken on trust — and the two designers can edit it there like any other design.

The same rotation is also supplied as proposal-import.txt (tab-separated, pastes directly) and proposal.json (the app's own format).

Alternatives measured

DESIGN	RUN	FF PRESENT	WKENDS	ONE-TURN	STEP	WK FIT	SCORE
WL4-24-F7 · f0d403d6 — Weekday Lates 4 (this proposal)	7	1	6	10/20	2h 34m	46.9	—
Same Turns · d15e1b74 — the searched proposal	6	1	6	18/20	2h 33m	58.5	—
By the Book · 0f14abce — the searched proposal	6	0	6	12/20	2h 52m	41.1	—
WL-24-EXT — Weekday Lates, before any change · a52ec588	9	5	6	7/20	4h 43m	60.8	—
WL3-24-F7 — the same five weeks, all MOVED · a6234195	6	1	4	12/20	2h 23m	46.9	—
WL2-24-R21 — no protected weeks at all · 33f70893	6	1	6	14/20	2h 45m	36.5	—
Block on lines 22–24 — one MORE weekend, weeks not on their own numbers · ede7e179	7	1	7	8/20	4h 6m	46.9	—
Block on lines 9–11 — FF19 also at its floor, one fewer weekend · 563b48ae	7	1	5	11/20	2h 54m	46.9	—
Workspace default · 29ec9810 (Dec 2026 table, generated)	7	5	5	0/20	4h 16m	41.1	—
Today's 20-line link (for scale)	9	5	4	7/16	4h 0m	57.7	—

How the winner was picked: rules first (rest, run, fatigue factors present, weekends off), then *Wk fit* — how evenly the weekday cover follows the

December traffic curve (lower is better; today's link scores what it scores). Two candidates tied on every rule; the fit decided it, and the runner-up is named so it can be asked for. Score is the search's own feel objective (lower is better) and is not a verdict. The workspace default is the app's own December table, generated and reordered with every switch on — a good design by every panel, and not one that resembles today's.

LOAD IT

The rotation, ready to paste

Line number, then Sunday to Saturday. SP is a cover week. Paste the whole block into Links → Import.

1	SP	SP	SP	SP	SP	SP	SP
2	08:30-16:30	RD	RD	06:20-14:20	06:20-14:20	06:20-14:20	06:20-14:20
3	RD	RD	08:30-16:30	08:30-16:30	07:00-15:30	07:00-15:30	06:20-14:50
4	RD	RD	14:00-22:30	13:30-22:00	14:00-22:30	14:00-22:30	14:30-22:00
5	14:30-23:25	RD	RD	13:30-22:00	13:30-22:00	13:30-22:00	13:30-22:00
6	13:30-22:00	13:30-22:00	RD	RD	13:30-22:00	13:00-21:00	14:45-23:55
7	SP	SP	SP	SP	SP	SP	SP
8	07:15-15:45	06:20-13:45	06:20-13:45	RD	RD	06:20-13:35	08:00-16:00
9	07:15-15:45	06:20-13:45	06:20-13:45	06:20-13:45	06:20-13:45	RD	RD
10	RD	06:20-14:20	07:00-15:30	07:00-15:30	06:20-14:20	06:20-14:20	06:20-13:45
11	RD	RD	06:20-14:20	12:30-21:00	12:30-21:00	12:30-21:00	RD
12	SP	SP	SP	SP	SP	SP	SP
13	10:30-19:00	12:30-21:00	12:30-21:00	RD	RD	12:30-21:00	13:30-21:00
14	14:30-23:25	14:00-22:30	RD	RD	RD	07:00-15:30	06:20-14:50
15	07:15-15:45	06:20-13:45	06:20-13:35	06:20-13:35	06:20-13:35	RD	RD
16	RD	08:30-17:00	08:30-17:00	08:30-17:00	RD	RD	RD
17	SP	SP	SP	SP	SP	SP	SP
18	14:30-23:25	16:25-23:55	RD	RD	13:30-22:00	16:25-23:55	RD
19	RD	14:00-22:30	14:00-22:30	14:00-22:30	14:00-22:30	14:00-22:30	RD
20	RD	07:00-15:30	07:00-15:30	06:20-13:45	07:00-15:30	06:20-13:45	06:20-14:50
21	RD	RD	16:25-23:55	16:25-23:55	16:25-23:55	16:25-23:55	16:25-23:55
22	RD	16:25-23:55	16:25-23:55	16:25-23:55	16:25-23:55	16:25-23:55	RD
23	RD	16:25-23:55	16:25-23:55	16:25-23:55	16:25-23:55	16:25-23:55	RD
24	RD	RD	13:30-22:00	13:30-22:00	13:30-22:00	RD	14:45-23:55

Prepared 22 September 2026. Every figure in this document was computed by the Marylebone Roster app's Links modules from exactly these cells; import them and the workspace's Design checks, hard limits, fatigue factors and coverage cards will restate them.